

A bounded pulse comparison: the released Navier–Stokes wave ODE and the coupled viscous pair

1 Scope and source equations

This note records one local, exact comparison. It does not identify the coupled Boussinesq stages with the released Navier–Stokes construction and makes no claim about a global Navier–Stokes solution. The source is the released PDF `openai_navier_stokes.pdf`, SHA-256 `8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81`. The source statements used here are Section 7, equations (7.2)–(7.8), (7.12)–(7.22), and Section 9, equations (9.1)–(9.2), on PDF pages 74–75, 77–81, and 101–103. In the source notation, on one lifted pulse rectangle,

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad n = \nabla_* \Phi, \quad (1)$$

with

$$n = (x_0 - v(pF_R + p_z G_R), p/R, p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (2)$$

For a nonzero integer harmonic m , the exact principal pulse equations are

$$n \cdot t_m = 0, \quad t'_m + \mathbf{K} t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad d_N = \varepsilon k^2 |n|^2, \quad (3)$$

where a prime is the pulse-coordinate derivative and

$$\mathbf{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + R F_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad \mathbf{A}_\Phi = -\mathbf{K} + \frac{n(n^T \mathbf{K} - (n')^T)}{|n|^2}. \quad (4)$$

The source pressure coefficient is exactly

$$\pi_m = \frac{i}{k m |n|^2} (n \cdot \mathbf{K} t_m - (n') \cdot t_m + n \cdot f_m). \quad (5)$$

For $m = 1$, a real homogeneous pulse t_1 has the source energy identity (7.22):

$$\frac{1}{2}(|t_1|^2)' = -g \cdot (t_{1,r}(t_{1,\theta}, t_{1,z})) - \varepsilon k^2 |n|^2 |t_1|^2, \quad (6)$$

where $g = (R F_R, G_R)$ in the tangential coordinates. Thus the source pulse retains both shear transfer and viscous dissipation.

2 The coupled physical pair

For one activated layer of the coupled calculation, put

$$a_c = -\frac{J\zeta \cdot G_{<q}}{Kr^2}, \quad b_c = K\zeta_1, \quad \lambda_c^\Theta = \kappa K^2 r^2, \quad \lambda_c^\Omega = \varepsilon_c K^2 r^2. \quad (7)$$

Its exact retained-viscosity pair, including optional scalar and vorticity controls, is

$$\dot{y} = C_c y + c_c, \quad y = \begin{pmatrix} \Theta \\ \Omega \end{pmatrix}, \quad C_c = \begin{pmatrix} -\lambda_c^\Theta & a_c \\ b_c & -\lambda_c^\Omega \end{pmatrix}, \quad c_c = \begin{pmatrix} c_\theta \\ c_\omega \end{pmatrix}. \quad (8)$$

Here $G_{<q}$, ζ , and r evolve with the inherited background and phase ODEs; in particular no fixed-background substitution is made. Let v be a source pulse coordinate on an interval and let $q_v = dt/dv$ describe its physical-time parametrization. The forward source parametrization has $q_v > 0$; the algebra below also retains a negative orientation without changing the amplitude sign. Define $y'(v) = q_v(C_c y + c_c)$. Here K, r are the coupled frequency and phase length, not the source carrier k and cylindrical radius R . The original A_0, λ_0, μ and time scale $\nu = \sqrt{A_0}$ are unchanged. The source chart parameter ε is not identified with either physical diffusivity.

3 Exact local residual map

Let $B(v) \in \mathbb{R}^{3 \times 2}$ be a C^1 tangent frame with full column rank and $n(v)^T B(v) = 0$, so its range is exactly $n(v)^\perp$. Let $B_\ell(v)$ be any left inverse, $B_\ell B = I_2$; then BB_ℓ acts as the identity on $n^\perp$. Let $L(v) \in \mathbb{R}^{2 \times 2}$ be any C^1 coefficient map and embed the coupled pair into a transverse source amplitude by

$$t(v) = B(v)L(v)y(v). \quad (9)$$

For an embedding with a non-tangent frame the transversality condition would be

$$n(v)^T B(v)L(v)y(v) = 0. \quad (10)$$

For the tangent frame used here, the condition holds for every L, y . Set

$$H(v) = B_\ell(v)(A_\Phi(v)B(v) - B'(v)). \quad (11)$$

Then the exact projected source residual of the embedded coupled pair is

$$\mathcal{E}_{L,c} := B_\ell \text{proj}_{n^\perp} [t' - A_\Phi t + m^2 d_N t + f_m] = [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \quad (12)$$

$$+ B_\ell \text{proj}_{n^\perp} f_m. \quad (13)$$

In particular, with $L = I$ and no added source, the defect is the explicit matrix expression

$$\mathcal{E}_{I,0} = (q_v C_c - H + m^2 d_N I_2) y. \quad (14)$$

This gives a local residual relation with every actual coefficient retained.

Proof. Differentiate (9): $t' = B'Ly + BL'y + BLy'$. Since $y' = q_v(C_c y + c_c)$, differentiate $n^T B = 0$ to obtain $n^T B' = -(n')^T B$. Also $n^T A_\Phi B = -(n')^T B$, directly from (4). Thus $A_\Phi B - B'$ has transverse range and equals BH . Applying B_ℓ to the transverse terms and $B_\ell \text{proj}_{n^\perp}$ to the force proves (13). More explicitly,

$$n^T(t' - A_\Phi t + m^2 d_N t) = 0, \quad n^T(t' - A_\Phi t + m^2 d_N t + f_m) = n^T f_m. \quad (15)$$

The unprojected force need not be transverse. The pressure below removes precisely this remaining normal component. $\square$

4 Pressure reconstruction and full three-dimensional residual

For any embedded transverse t and arbitrary f_m , define

$$\pi_{L,c} = \frac{i}{km|n|^2} (n \cdot \mathbf{K}t - (n') \cdot t + n \cdot f_m). \quad (16)$$

Then the full source residual satisfies the exact decomposition

$$\mathcal{R}_{L,c} := t' + \mathbf{K}t + m^2 d_N t + ikmn\pi_{L,c} + f_m \quad (17)$$

$$= \text{proj}_{n^\perp} (t' + \mathbf{K}t + m^2 d_N t + f_m) = B \mathcal{E}_{L,c}, \quad (18)$$

where the last equality uses the definition of H and the transverse condition. Thus the pressure removes precisely the normal component; it does not remove $\mathcal{E}_{L,c}$, the moving-frame, shear, diffusion, or coupled-background mismatch. If one elects to force the source pulse to obey the released equation, then the unique additional transverse forcing is

$$f_{m,\text{corr}} = -B \mathcal{E}_{L,c}, \quad (19)$$

added to the old f_m . The old $\mathcal{E}_{L,c}$ is evaluated before this addition, so the new projected residual is zero. Since $n^T f_{m,\text{corr}} = 0$, pressure is unchanged. This is an exact local forced relation, not an identification of the physical constructions.

5 Diffusion and energy comparison

The source diffusion multiplier is $m^2 d_N = m^2 \varepsilon k^2 |n|^2$ in (3). The coupled pair has the two retained physical losses λ_c^Θ and λ_c^Ω , which are generally unequal to $m^2 d_N$. Their signed discrepancy in (14) is explicitly

$$(m^2 d_N - q_v \lambda_c^\Theta) \Theta, \quad (m^2 d_N - q_v \lambda_c^\Omega) \Omega \quad (20)$$

when $H = 0$ and C_c is expanded entrywise; with general H the exact diagonal entries are those of $q_v C_c - H + m^2 d_N I_2$. No damping is dropped. If one compares nonnegative loss magnitudes instead, the corresponding cost gaps are $|m^2 d_N - q_v \lambda_c^\Theta|$ and $|m^2 d_N - q_v \lambda_c^\Omega|$. The required finite-interval coordinate map can be constructed explicitly. Let $I = [v_0, v_1]$, and let X_c, X_s be the identity-initial-value fundamental matrices for $D_c = q_v C_c$ and $D_s = H - m^2 d_N I_2$, respectively. For either continuous matrix D , define

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{v_0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

With $M_D = \sup_I \|D\|$ and $T_I = v_1 - v_0$, the j th integral has norm at most $(M_D T_I)^j / j!$. Uniform convergence proves the integral equation $X_D = I_2 + \int_{v_0}^v D X_D$ and hence $X_D' = D X_D$. The same iteration for $Z' = -ZD$, $Z(v_0) = I_2$ gives $Z X_D = I_2$ by differentiation, proving invertibility on I . For a specified invertible L_0 , set

$$L = X_s L_0 X_c^{-1}, \quad L^{-1} = X_c L_0^{-1} X_s^{-1}.$$

This has norm at most $\|L_0\| e^{(M_c + M_s) T_I}$; its inverse has the same bound with L_0^{-1} . Differentiation proves

$$L' = H L - q_v L C_c - m^2 d_N L, \quad (21)$$

so the homogeneous embedded pair has zero projected residual. This proves an actual finite-interval conjugacy along the retained coefficient path. It does not make the source phase, pressure, cutoff, or nonlinear covariance equal to the coupled Boussinesq fields. For general controls c_c , the forced term is $q_v L c_c$ in (13).

For an arbitrary embedded field (possibly complex), the exact energy balance retains its full defect:

$$\begin{aligned} \frac{1}{2}(|t|^2)' = & -\operatorname{Re}(\bar{t}^T \mathbf{K} t) - m^2 d_N |t|^2 - \operatorname{Re}(\bar{t}^T f_m) \\ & + \operatorname{Re}(\bar{t}^T B \mathcal{E}_{L,c}). \end{aligned} \quad (22)$$

This follows by taking the real Hermitian product of (18) with t . Its pressure term vanishes because n is real and $n^T t = 0$. The $2F$ part of $\mathbf{K}$ is skew-symmetric, hence its real work is zero; the shear work equals $\operatorname{Re}(\bar{t}_r(g_\theta t_\theta + g_z t_z))$. Only for a zero residual, zero force, $m = 1$, and real t does this reduce to (6). The real coupled pair has

$$\frac{1}{2} \frac{d}{dt} |y|^2 = -\lambda_c^\Theta \Theta^2 - \lambda_c^\Omega \Omega^2 + (a_c + b_c) \Theta \Omega + c_\theta \Theta + c_\omega \Omega, \quad (23)$$

which is a two-component physical energy identity. Equations (13) and (23) exhibit exactly where the source's three-dimensional shear flux and the coupled pair's two-dimensional exchange differ.

6 Proven scope

The source uses a cylindrical, localized, divergence-free curl construction (Section 7.4 and Section 9, PDF pages 85–87 and 102–103); the displayed principal pulse equation is only one component of that construction. The comparison above proves an exact local map and its full pressure-projected residual for every finite interval on which the denominators and frame left inverse exist. It preserves source viscosity, moving phase normal, pressure, rotation/shear matrix, and all coupled physical diffusion coefficients. It does not prove that the coupled third-growth or steering stages produce the source's pulse phase Φ , covariance, cutoff tails, or smooth terminal force. Those remain separate bridge problems.

The sign conventions in this comparison are also explicit. The quantities $q_v > 0$, $\varepsilon > 0$, and $d_N = \varepsilon k^2 |n|^2 > 0$ record the chosen orientation of physical time and the positive viscous coefficient; they are not assumptions on the sign of a pulse or blowout amplitude. The equations (3)–(5) are linear in (t_m, f_m) , so the simultaneous replacement $(t_m, f_m, \pi_m) \mapsto (-t_m, -f_m, -\pi_m)$ is an exact reflected branch. Likewise, the local map (9) is sign-equivariant under $y \mapsto -y$ and $c_c \mapsto -c_c$, with the coefficients fixed at the actual path. This is a statement about the retained linear equations, not a sign symmetry of the entire coupled nonlinear system. The source covariance is quadratic, so reversing a wave leaves that covariance unchanged. Reversing the full Navier–Stokes velocity need not reverse its force, since $(u \cdot \nabla)u$ is even in u . The full-flow map that reverses axisymmetric swirl is a spatial reflection; its complete formula is proved in the accompanying covariance bridge. No sign of a blowout or endpoint is inferred from positive geometric scales.